

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/01

Paper 1 Multiple Choice

30 August 2016

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

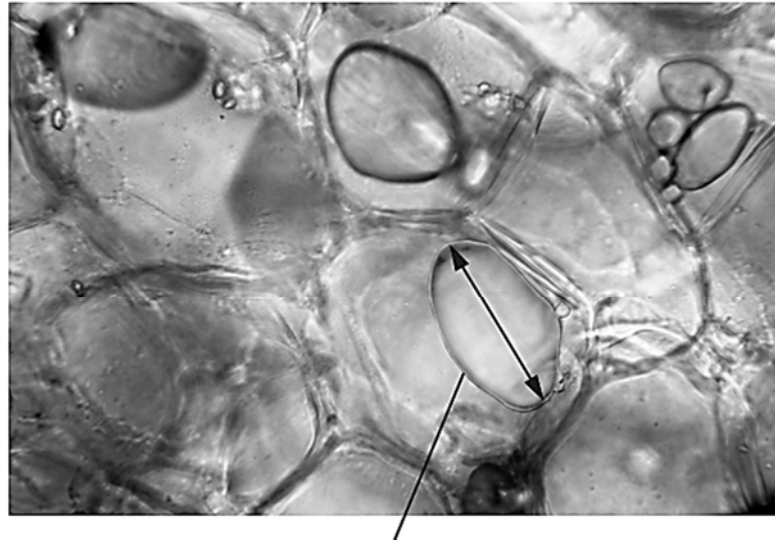
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **22** printed pages.



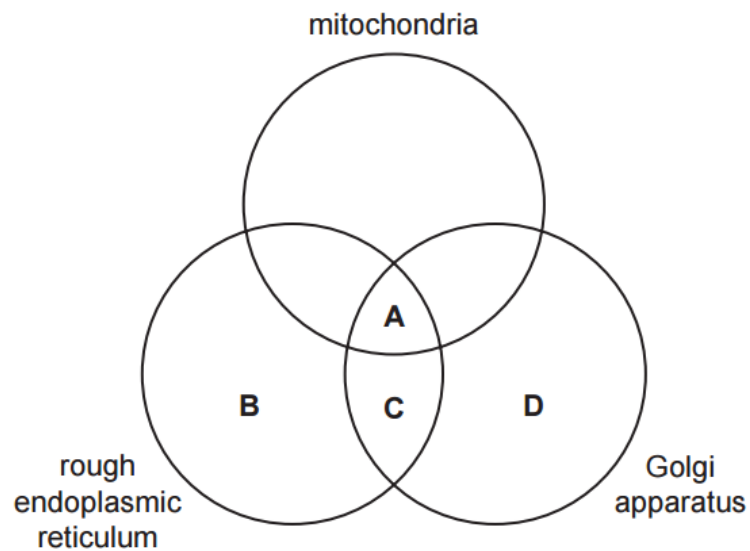
- 1 The picture shows starch grains as seen with an optical microscope. The actual length of the starch grain A is $50\text{ }\mu\text{m}$.



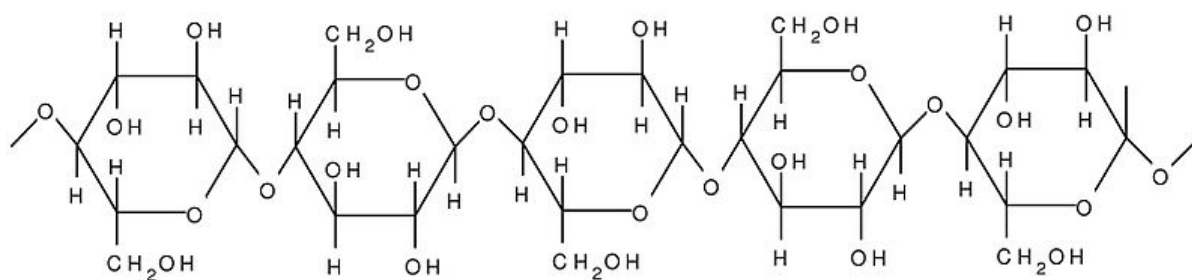
Starch grain A

Calculate the magnification.

- A x 5
 B x 50
 C x 500
 D x 5000
- 2 Which of the following organelle(s) is/are directly required for the formation of the hydrolytic enzymes found in lysosomes?

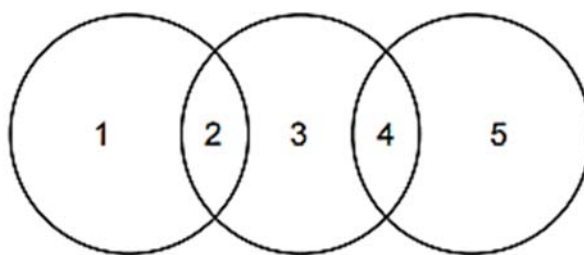


- 3 The figure below shows a portion of a polymer.



Which statement is **true** about the polymer?

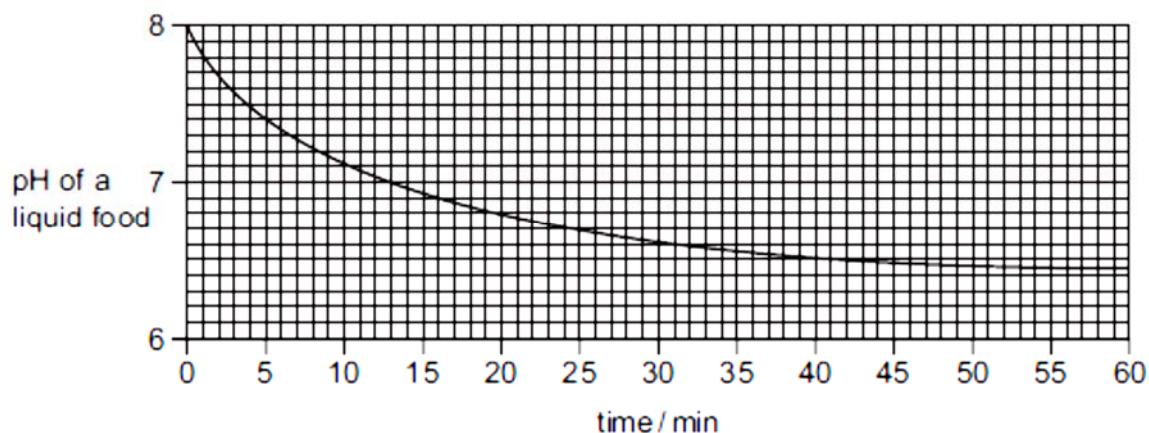
- A The polymer will assume a helical structure.
 - B The polymer can exist in both branched and unbranched forms.
 - C The orientation of the monomer will result in a straight chain polymer.
 - D The monomers are able to form α (1-6) glycosidic bonds with one another.
- 4 The diagram shows the relationship between the levels of protein structure and bonds.



Which row is **correct**?

	1	2	3	4	5
A	primary	peptide	secondary	ionic	tertiary
B	secondary	hydrogen	tertiary	peptide	primary
C	tertiary	ionic	primary	peptide	quaternary
D	quaternary	ionic	tertiary	ionic	secondary

- 5 Lipase is a digestive enzyme produced by the pancreas that catalyses the hydrolysis of dietary lipids. The table shows how the pH of a liquid food containing a high proportion of lipids decreases over time.



Which of the following statements are possible explanations of the results of the experiment between 50 and 60 minutes?

- 1 Enzyme concentration becomes the limiting factor.
 - 2 Substrate concentration becomes the limiting factor.
 - 3 All the enzyme active sites are saturated.
 - 4 Denaturation of the enzyme by the products.
 - 5 Products are acting as inhibitors.
- A** 1, 2 and 3
B 1, 4 and 5
C 2, 3 and 4
D 2, 4 and 5

6 Specific enzyme inhibitors inhibit only one enzyme.

The drug disulfiram, which is used as a treatment for alcoholism, is a specific inhibitor of acetaldehyde dehydrogenase. Acetaldehyde dehydrogenase is involved in the detoxification of ethanol. As a result of inhibition by disulfiram, any ethanol that is present in the system can only be partly broken down, resulting in nausea and vomiting.

Why is it important that the enzyme inhibitor disulfiram is specific?

- 1 It cannot disrupt other metabolic pathways.
- 2 It prevents ethanol from binding to the active site.
- 3 It is unlikely to cause unwanted side effects.
- 4 It inhibits aldehyde oxidase.

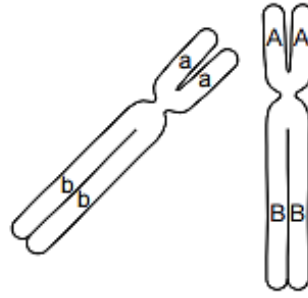
- A** 1, 2, 3 and 4
B 1, 2 and 3 only
C 1 and 3 only
D 2 and 4 only

7 In a multicellular organism, which of these statements about mitosis can help to explain the control of the mitotic cell cycle?

- 1 In most cells the genes initiating mitosis are not switched on.
- 2 Mitosis produces cells to replace damaged cells that cannot be repaired.
- 3 Mitosis transmits a complete copy of all the alleles in a cell to new cells.
- 4 Daughter cells formed by asexual reproduction develop from unspecialised cells.

- A** 1 only
B 3 only
C 1 and 4
D 2 and 4

- 8 The diagram shows two homologous chromosomes in early prophase I of meiosis in an animal cell. Two genes, A/a and B/b, whose loci occur on the homologous chromosomes are also shown.



Which row of diagrams is a possible representation of these chromosomes as they progress from anaphase I to prophase II?

	anaphase I	prophase II
A		
B		
C		
D		

- 9 The nucleic acids present in a cell of the bacterium *Escherichia coli* were analysed. Some of the results are shown in the table.

type of nucleic acid	number of different variants	number of molecules per cell	percentage of dry mass of cell
1	600	2500	2
2	60	160 000	3
3	2	20 000	21

Which row identifies each type of nucleic acid 1, 2 and 3?

	1	2	3
A	DNA	mRNA	tRNA
B	mRNA	tRNA	rRNA
C	rRNA	DNA	mRNA
D	tRNA	rRNA	DNA

- 10 How many of these statements about DNA polymerases are **correct**?

- 1 They transcribe DNA.
- 2 They synthesise DNA in the 3' to 5' direction.
- 3 They require a primer to function.
- 4 They require activated nucleotides.

- A 1
B 2
C 3
D 4

- 11 What is the function of the enzyme RNA polymerase?

- A to form a polypeptide using mRNA as a template
B to form a strand of DNA using mRNA as a template
C to form a strand of mRNA using DNA as a template
D to form a strand of mRNA using tRNA as a template

- 12** The following sequence of bases shows a short section of linear DNA from which mRNA is transcribed.

TACTCACATTAG...

The table shows a number of mRNA codons and their corresponding amino acids.

codon	AGU	AUC	AUG	CAU	GUA	UAC	UAG	UCA
amino acid	serine	iso-leucine	methionine	histidine	valine	tyrosine	'stop'	serine

Which row shows how this short section of linear DNA would be translated into part of a polypeptide chain?

	tRNA anti-codon order	amino acid sequence
A	AUGAGUGUAAUC	methionine, serine, valine, iso-leucine
B	AUGAGUGUAAUC	tyrosine, serine, histidine, stop
C	UACUCACAUUAG	methionine, serine, valine, iso-leucine
D	UACUCACAUUAG	tyrosine, serine, histidine, stop

- 13** About 20% of all human pregnancies are estimated to be lost by spontaneous abortion (miscarriage). About half of these spontaneous abortions are associated with chromosome aberrations.

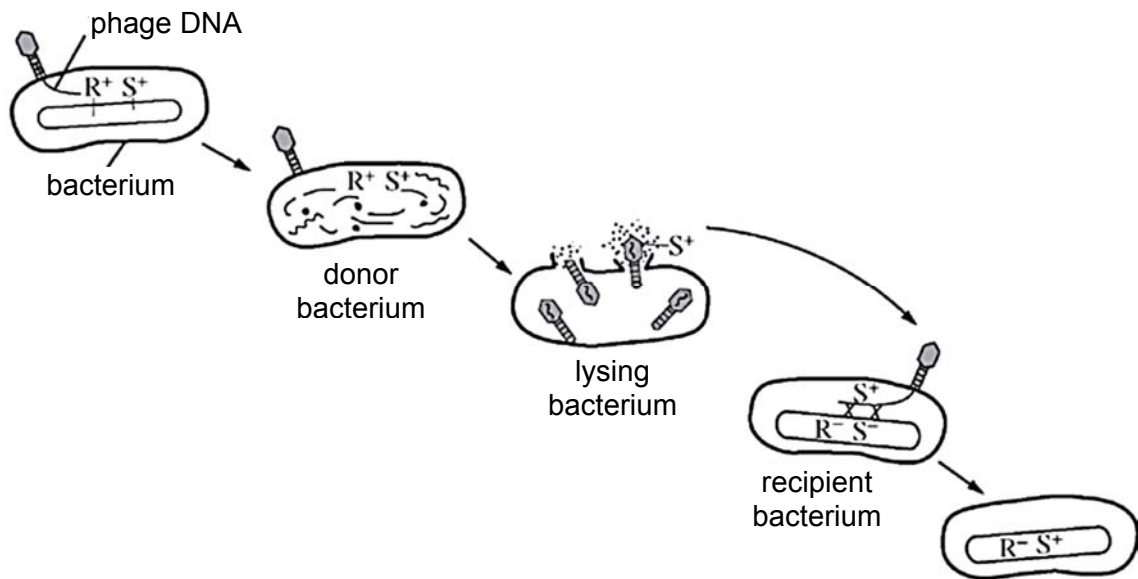
The table shows the percentage incidence of each type of chromosome aberration that was detected in spontaneous abortions associated with chromosome aberrations.

chromosome aberration	percentage incidence
trisomy ($2n+1$)	62
monosomy ($2n-1$)	18
triploidy ($3n$)	17
translocations	3

What may be concluded from these data?

- 1 Numerical aberrations are much more likely to be associated with spontaneous abortions than are structural aberrations.
 - 2 The occurrence of three sets of chromosomes has a similar percentage incidence as monosomy.
 - 3 The presence of an extra chromosome is more likely to be associated with spontaneous abortion than is the lack of one chromosome.
- A** 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

- 14 The diagram below shows several steps in the gene transfer process between bacteria.



Which of the following statements explains how genetic variation in a population of bacteria may result from this process?

- A Bacterial proteins are transferred from the donor bacterium to the recipient bacterium and recombine with the chromosome.
- B The recipient bacterium incorporates the transduced genetic material into its own chromosome and synthesizes the corresponding proteins.
- C The phage infection of the recipient bacterium and the introduction of the donor's DNA caused random mutations in the recipient's chromosome.
- D DNA of the recipient's chromosome undergoes recombination with the donor's DNA, leading to a change in its genotype.

- 15** The only cells that an enveloped virus can infect are those whose cell surface membranes have specific receptors complementary to proteins in the virus envelope. The virus enters the cell by a type of endocytosis involving the protein clathrin, which is produced in the host cell.

The following are some of the events that follow attachment of an enveloped virus to a cell surface membrane.

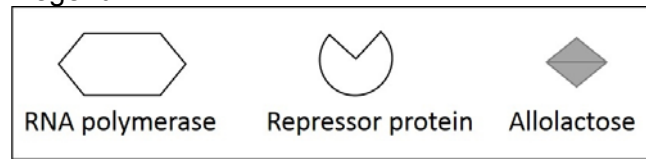
- 1 A vesicle containing a virus fuses with an endosome (cytoplasmic vesicle).
- 2 The virus is surrounded by its envelope, a membrane and a layer of clathrin.
- 3 An acid pH causes the viral envelope to fuse with the membrane of an endosome.
- 4 The protein clathrin produces a depression in a cell surface membrane.
- 5 Clathrin is removed and recycled.

What is the correct sequence of these events in order to leave the virus free of its envelope in the cytoplasm of an infected cell?

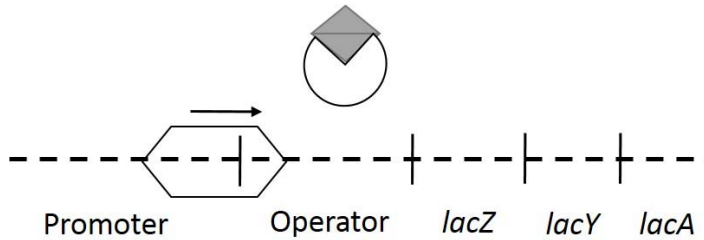
- A** 2 → 3 → 1 → 4 → 5
B 2 → 4 → 5 → 3 → 1
C 4 → 1 → 3 → 2 → 5
D 4 → 2 → 5 → 1 → 3

- 16 Using the legend provided, which of the following correctly depicts the interactions of the components at the *lac* operon when lactose is absent from the medium?

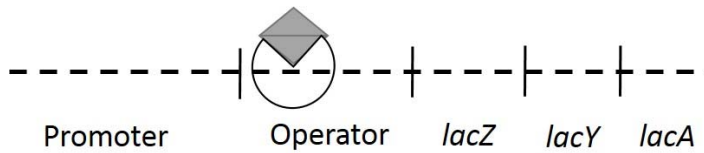
Legend:



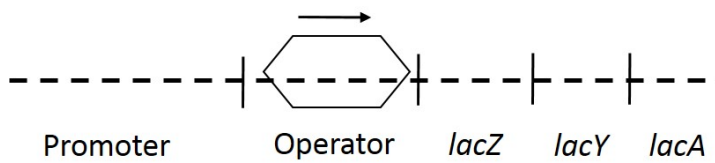
A



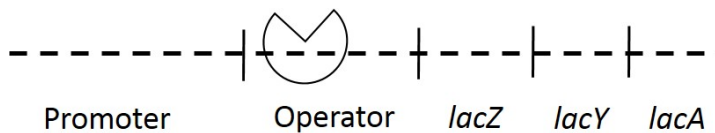
B



C



D



- 17 A type of bacteria caused fatalities in human. These strains of bacteria possess genes for a toxin not found in the other harmless strains.

In an attempt to find out how these genes can be transferred between bacteria, several experimental set-ups were carried out. The results are shown in the table below.

conditions	results
DNA isolated from virulent strain and incubated with harmless strain.	Some virulent strains observed.
Virulent and harmless strains of bacteria incubated in a container with no barrier.	Some virulent strains observed.
Virulent and harmless strains of bacteria incubated in a container but separated by physical membrane barrier.	No virulent strain found in the side with harmless bacteria.

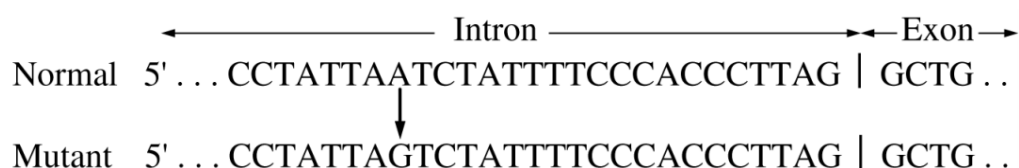
From the information provided only, which of the following gene transfer processes could have taken place?

- 1 transformation
- 2 conjugation
- 3 transduction

- A 1 only
 B 2 only
 C 1 and 2
 D 1 and 3

- 18 Portions of the DNA sequences and mutant β -globin genes are shown below.

The arrow ($\downarrow$) indicates the single base substitution that occurred to result in the disease β -thalassaemia.



The most plausible explanation for the effects of the mutation is that

- A there was a change in the codon that affected the amino acid coded for.
 B a recognition site for a restriction enzyme was generated, resulting in a DNA break.
 C a nonsense mutation occurred to result in a truncated polypeptide.
 D a new splice site was created, such that a portion of the intron was not removed.

- 19 One form of post-translational modification of a protein is the
- A methylation of the CpG islands.
 - B shuffling of exons to produce many types of mRNA from a single gene.
 - C removal of introns from the pre-mRNA.
 - D removal or modification of amino acids in the polypeptide.
- 20 Laboratory mice whose *p53* genes had been switched off developed tumours.

When their *p53* genes were switched on again, the tumour cells stopped dividing and died within a few days. Healthy cells in the mice were unaffected.

What do these observations suggest?

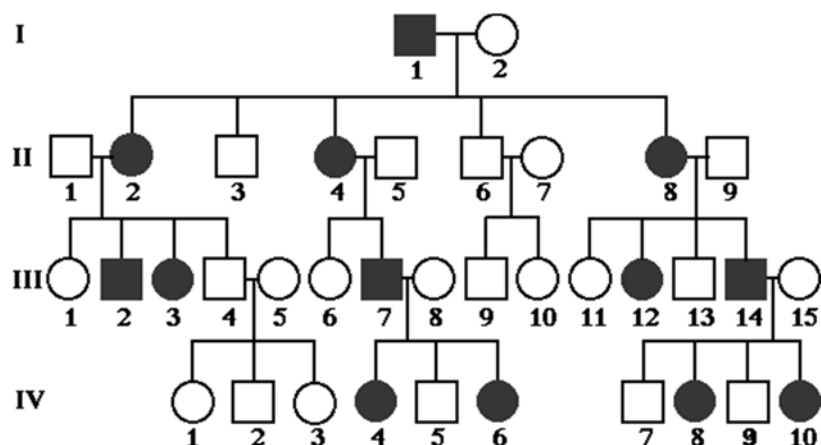
- A *p53* protein speeds up the mitotic cell cycle
 - B *p53* protein causes all cells to die
 - C the *p53* gene acts as a tumour suppressor gene
 - D the *p53* gene encourages the growth of tumours
- 21 In the breeding season, male Anole lizards court females by bobbing their heads up and down while displaying a colourful throat patch. Both characteristics are controlled by genes found on separate chromosomes. Anoles prefer to mate with lizards, which bob their heads fast and have red throat patches. These two alleles are dominant over their counterparts, slow bobbing and yellow throats.

A male lizard heterozygous for head bobbing and homozygous dominant for the red throat patch mates with a female that is also heterozygous for head bobbing but has yellow throat patch.

What percentage of the offspring has fast bobbing and red throat phenotype?

- A 25%
- B 50%
- C 75%
- D 100%

- 22 The inheritance of a genetic disease in a family is presented in a pedigree tree below.



What is the **most likely** type of inheritance shown?

- A autosomal dominant
 - B autosomal recessive
 - C sex-linked dominant
 - D sex-linked recessive
- 23 In a plant, three genes are known to be linked. The table below gives the recombination frequencies obtained from crosses involving pairs of these genes.

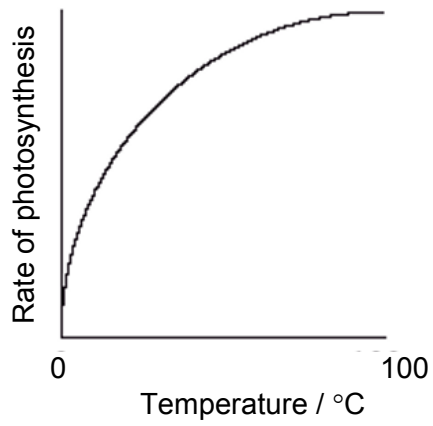
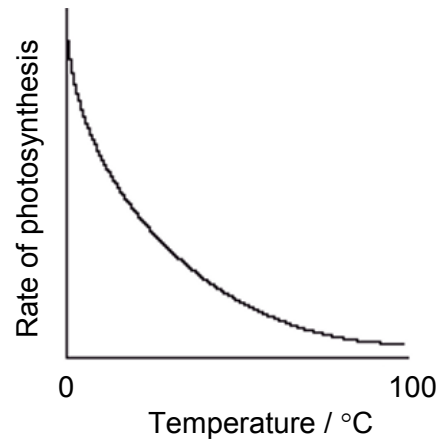
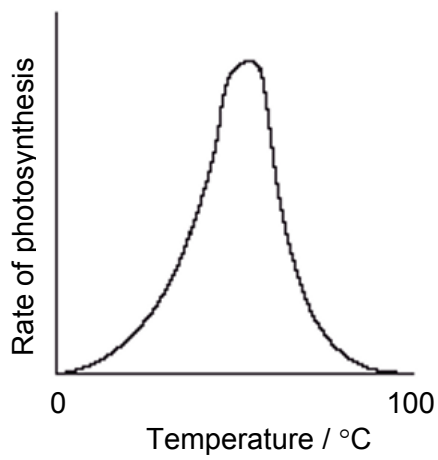
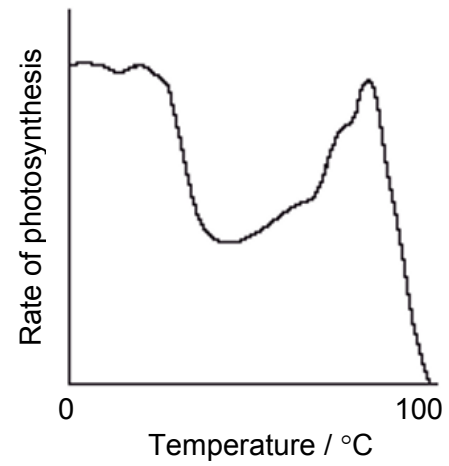
pair of linked genes	recombination frequency
colour of flowers (P) x appearance of fruit (Q)	0.43
appearance of fruit (Q) x presence of prickles (R)	0.17
presence of prickles (R) x colour of flowers (P)	0.26

- 1 The chance of crossing over occurring between genes P and R will be lower than that of Q and R.
- 2 Genes Q and R are 17 map units apart on the same chromosome.
- 3 The order of the genes on the chromosome is P – R – Q.
- 4 The recombination frequencies are obtained based on the proportion of recombinant offspring.

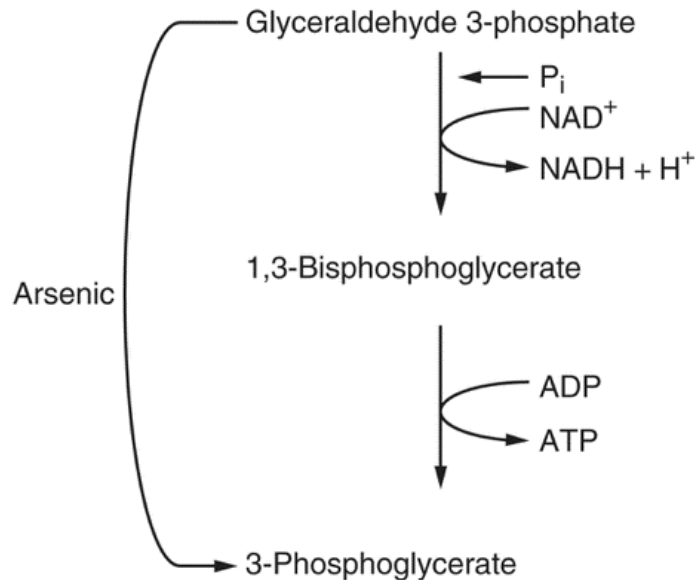
Based on the information, which of the above statement(s) is/are **false**?

- A 1 only
- B 1 and 2
- C 2 and 3
- D 2 and 4

- 24 Which graph best represents the effect of temperature on the rate of photosynthesis of a plant?

A**B****C****D**

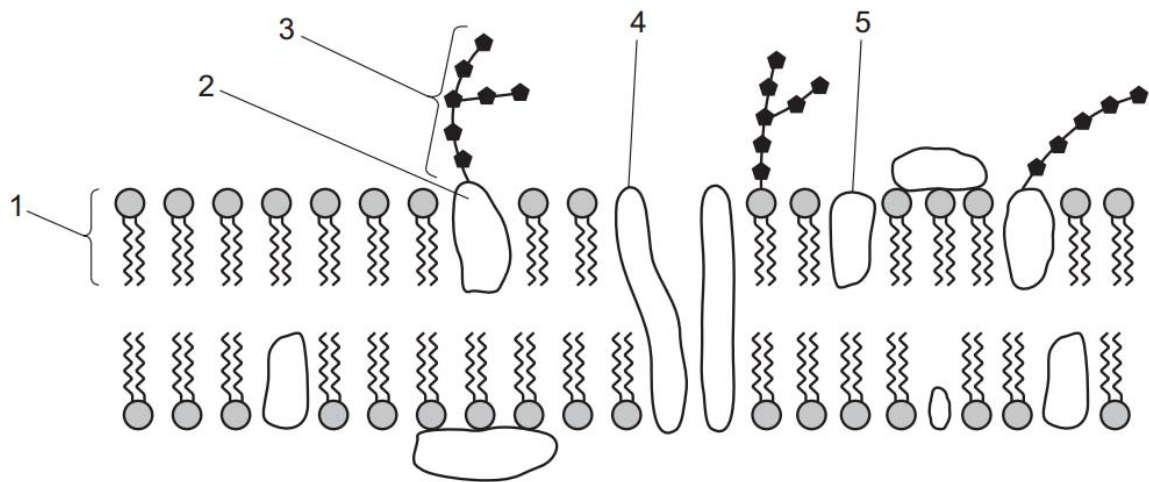
- 25 The diagram shows the effect of arsenic on the metabolism of glyceraldehyde-3-phosphate.



What is the net yield of ATP molecules from the glycolysis process involving 2 molecules of glucose in the presence of arsenic?

- A 0
 - B 1
 - C 2
 - D 4
- 26 For an action potential to occur,
- A Na^+ influx must exceed K^+ efflux.
 - B the membrane must not be in the relative refractory period.
 - C the stimulus must result in the exocytosis of neurotransmitters.
 - D voltage-gated potassium channels must be closed.
- 27 Caffeine is an inhibitor of phosphodiesterase. Therefore, the cells of a person who has recently consumed coffee would have increased levels of
- A ATP
 - B GTP
 - C AMP
 - D cAMP

28 The diagram shows part of a cell surface membrane.

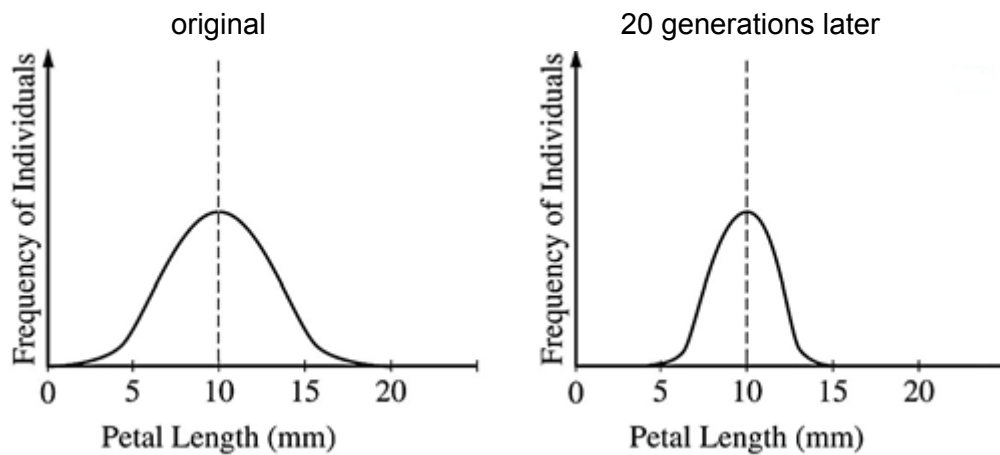


Which molecule(s) allow(s) the movement of hydrophilic substances across the membrane?

- A 1 only
- B 4 only
- C 2 and 5
- D 4 and 5

29 A plant species arrives at a new island and are exposed to a new set of pollinators.

The diagram below shows the frequency distribution of petal length in the original colonising population and 20 generations later.



Which type of selection is shown in this example?

- A directional selection
- B disruptive selection
- C neutral selection
- D stabilising selection

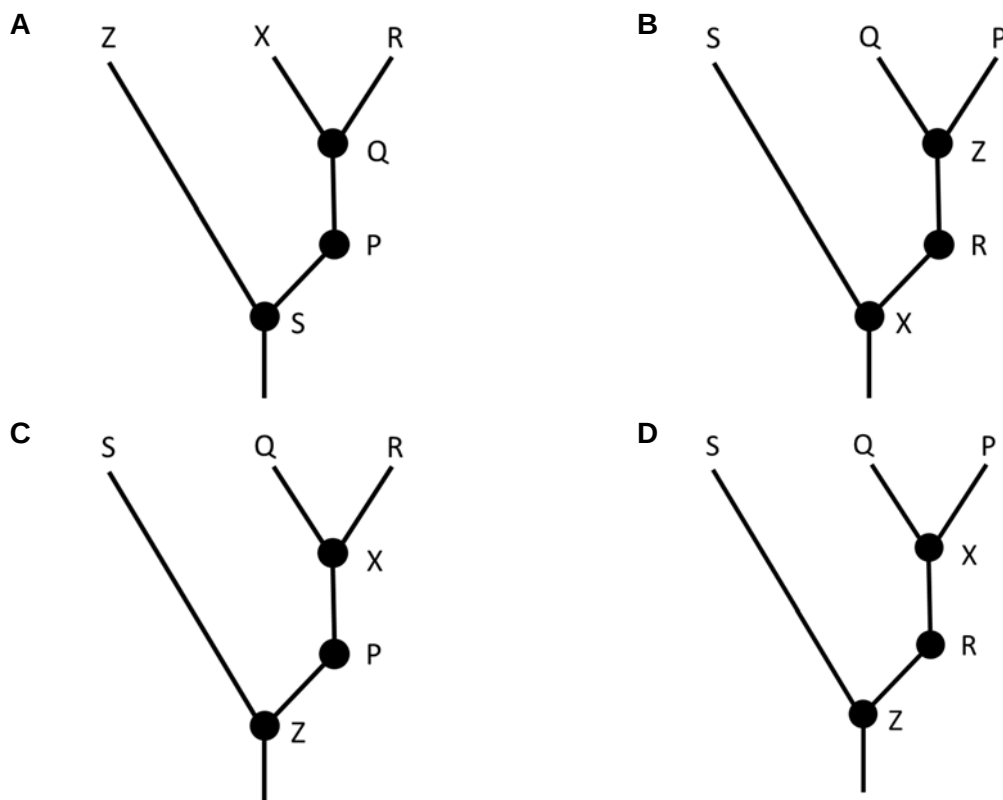
- 30** Q, P, R and S are related species of organisms.

Species X is an extinct recent common ancestor of species Q and R.

X, Q and R all evolved from species P.

Species S is the least related to the others, with extinct species Z being its most recent phylogenetic link to the other species.

Which of the following phylogenetic trees correctly represents the relationships described above?



- 31** When considering the neutral theory of molecular evolution, which of the following is a correct assumption of this model?
- A** New alleles that confer a higher fitness level tend to increase in frequency over time in the population.
 - B** New alleles that confer a lower fitness level tend to decrease in frequency over time in the population.
 - C** New alleles that confer an advantage to the heterozygotes are maintained in the population.
 - D** New alleles that do not confer any effects on the fitness level tend to fluctuate in the population randomly.

- 32** The human protein hormone somatotrophin can be produced by genetic engineering using plasmid vectors.

Which of the following statements describe the advantages of using such plasmids?

- 1 A gene coding for somatotrophin from a cDNA library can be inserted into the plasmid.
- 2 The plasmid carrying the gene coding for somatotrophin could contain the promoter sequence of DNA that is found in human cells expressing the hormone.
- 3 A synthetic gene produced from knowledge of the amino acid sequence of somatotrophin can be inserted into the plasmid.

- A** 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 3 only

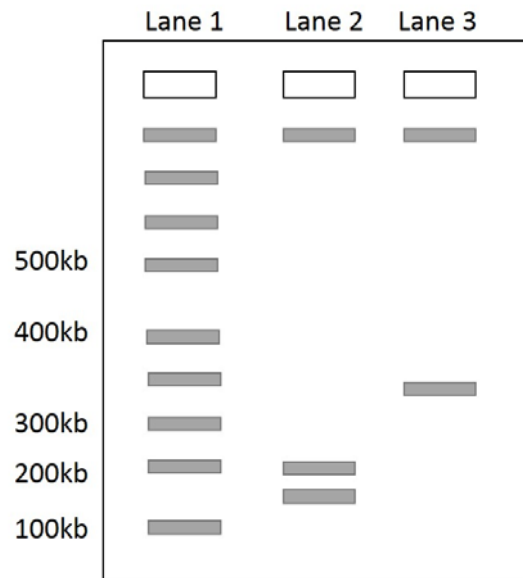
- 33** A polymerase chain reaction (PCR) amplification was performed on human genomic DNA.

Multiple products of varying sizes were obtained, including one of the expected size.

Which of the following modifications to the protocol is the most likely to eliminate the extra PCR products?

- A** increasing length of the primer
B increasing the denaturation temperature from 94°C to 96°C
C decreasing the annealing temperature from 56°C to 52°C
D increasing the elongation temperature from 70°C to 74°C

- 34 The diagram below shows the results from a restriction digestion of a normal *CFTR* allele (lane 2) and a mutant *CFTR* allele (lane 3). Lane 1 contains the DNA ladder.



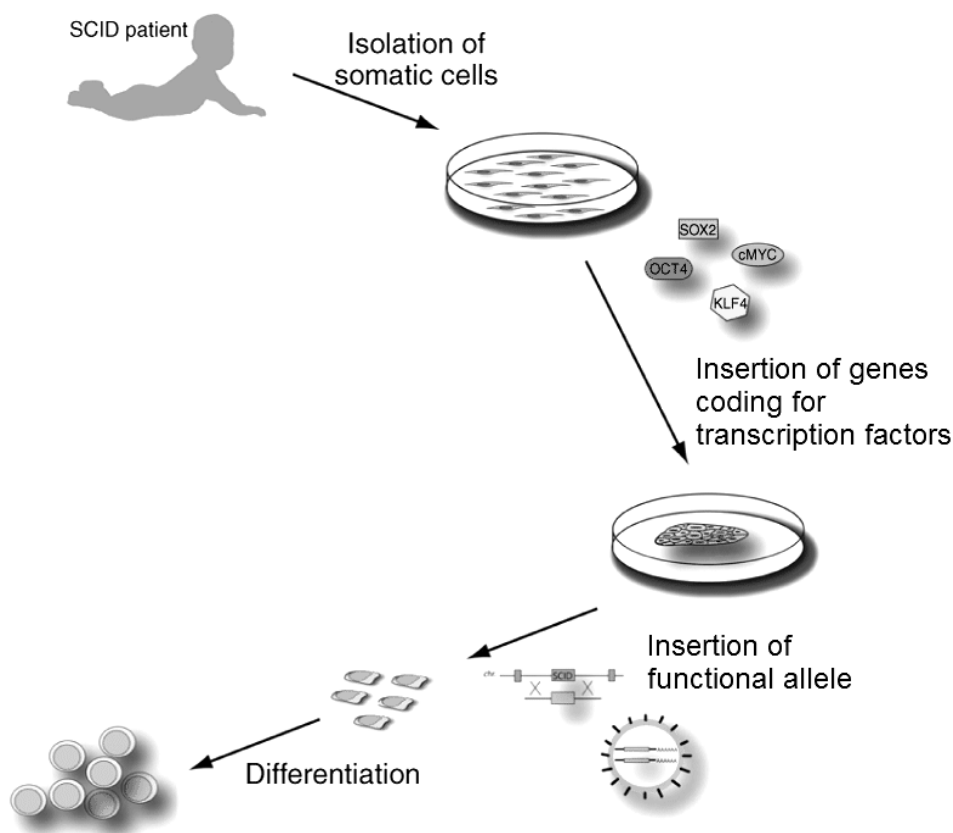
Based on the results shown, which of the following statements best describes the mutation that resulted in the mutant *CFTR* allele?

- A An additional restriction site was generated.
 - B One restriction site was lost.
 - C Two restriction sites were lost.
 - D The number of restriction sites remained unchanged.
- 35 Within its own environment, a particular cell line cannot be induced to produce a cell from a different cell line.

Which statement explains this?

- A Genes not required for a particular cell line are methylated.
- B Genes not required for a particular cell line are removed by enzymes.
- C Only pre-mRNA that is required for a particular cell line is processed.
- D Stem cells have only the genes required for their particular cell line.

- 36** In 2006, scientists have discovered a means to convert somatic cells, such as skin cells, into pluripotent stem cells. By inserting four specific genes which code for transcription factors, somatic cells are able to de-differentiate and achieve pluripotency. Such pluripotent stem cells may one day be used for gene therapy to treat diseases such as severe combined immunodeficiency (SCID), as shown in the following diagram.



Which statement shows one benefit of using these novel pluripotent stem cells over embryonic stem cells?

- A** A functional copy of the allele could be inserted into these novel stem cells using a retroviral vector during gene therapy.
- B** These novel stem cells have the potential to differentiate into more types of cells, hence they can be used to treat a greater variety of diseases.
- C** These novel stem cells will not be rejected by the host after being transplanted back into the body.
- D** These novel stem cells are able to maintain the undifferentiated state for long periods of time due to the insertion of additional genes.

- 37** Research into gene therapy that targets germ cells (sperm and ova) has raised some concerns.
- 1 The inserted gene may interfere with fetal development in unexpected ways.
 - 2 Developing babies cannot choose to have gene therapy.
 - 3 The inserted gene could prevent the inheritance of genetic diseases.

Which concerns are ethical reasons against the use of germ cell gene therapy?

- A** 1, 2 and 3
 - B** 1 and 2 only
 - C** 1 and 3 only
 - D** 2 and 3 only
- 38** In a callus culture,
- A** a high cytokinin to auxin ratio results in root formation.
 - B** a low cytokinin to auxin ratio results in root formation.
 - C** only auxin is required to induce shoot and root formation.
 - D** only cytokinin is required to induce shoot and root formation.
- 39** Why can improved varieties of plants and animals, used as human food, be developed more quickly by genetic engineering than by traditional selective breeding methods?
- A** Genetically engineered organisms mature more quickly and breed sooner.
 - B** The existing desirable features of the plants or animals are mostly unchanged.
 - C** The introduced genes can improve the quality of the food produced.
 - D** The organisms used for genetic engineering are already used for food.
- 40** Which statement supports the view that genetically engineered animals could help to solve the demand for food in the world?
- A** Transgenic pigs and sheep are produced to express higher levels of growth hormone.
 - B** Biomedical applications of genetically engineered animals have also become routine within the pharmaceutical industry, for drug discovery, drug development and risk assessment.
 - C** Cloning of either extinct or endangered species such as thylacine and woolly mammoth helps to retain genetic diversity in small populations.
 - D** By inserting genes from sea anemone and jellyfish, zebrafish have been genetically engineered to express fluorescent proteins.